

```
Processando config 12, seed 3, teste: Rabbit
XXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXX
Starting Rabbit: nb = 16777216
Version: TestU01 1.2.3
XXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXX
```

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*****
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```
HOST = matheus-IdeaPad-3-15ALC6, Linux
```

```
ufile_CreateReadBin: /home/matheus/Área de
Trabalho/qanalog_prng-main/qanalog_prng-main/outputs/out_config12_seed3.bin
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```
smultin_MultinomialBitsOver test:
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```
N = 2, n = 8388608, r = 0, s = 32, L = 38,
Sparse = TRUE
```

```
Number of bits = n = 8388608
Number of cells =  $2^L$  = 274877906944
Expected number per cell =  $1 / 32768$ 
EColl =  $n^2 / (2k)$  = 128
Hashing = TRUE
```

```
Collision test
CollisionOver: density =  $n / k = 1 / 32768.00$ 
Expected number of collisions =  $\mu = 128.00$ 
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```
Results of CollisionOver test:
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POISSON approximation :
Expected number of collisions =  $N \cdot \mu$  : 256.00
Observed number of collisions : 281
p-value of test : 0.06
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```
Total number of cells containing j balls
```

```
j = 0 : 549739036953
j = 1 : 16776654
j = 2 : 281
j = 3 : 0
j = 4 : 0
j = 5 : 0
```

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```
CPU time used : 00:00:04.44
```

```
Generator state:
16777216 bits have been read.
```

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*****
```

```
HOST = matheus-IdeaPad-3-15ALC6, Linux
```

```
ufile_CreateReadBin: /home/matheus/Área de
Trabalho/qanalog_prng-main/qanalog_prng-main/outputs/out_config12_seed3.bin
```

snpair_ClosePairsBitMatch test:

N = 1, n = 262144, r = 0, t = 2

Max of all bit distances : 16.00
p-value of test : 0.86

CPU time used : 00:00:00.05

Generator state:
33554432 bits have been read.

HOST = matheus-IdeaPad-3-15ALC6, Linux

ufile_CreateReadBin: /home/matheus/Área de
Trabalho/qanalog_prng-main/qanalog_prng-main/outputs/out_config12_seed3.bin

snpair_ClosePairsBitMatch test:

N = 1, n = 131072, r = 0, t = 4

Max of all bit distances : 7.00
p-value of test : 0.86

CPU time used : 00:00:00.07

Generator state:
50331648 bits have been read.

HOST = matheus-IdeaPad-3-15ALC6, Linux

ufile_CreateReadBin: /home/matheus/Área de
Trabalho/qanalog_prng-main/qanalog_prng-main/outputs/out_config12_seed3.bin

svaria_AppearanceSpacings test:

N = 1, Q = 524288, K = 524288, r = 0, s = 30, L = 15

Sequences of $n = (K + Q)L = 15728640$ bits
Q = 524288 initialization blocks
K = 524288 blocks for the test
the blocks have L = 15 bits

Normal statistic : -0.36
p-value of test : 0.64

CPU time used : 00:00:00.01

Generator state:
67108864 bits have been read.

HOST = matheus-IdeaPad-3-15ALC6, Linux

ufile_CreateReadBin: /home/matheus/Área de
Trabalho/qanalog_prng-main/qanalog_prng-main/outputs/out_config12_seed3.bin

scomp_LinearComp test:

N = 1, n = 18176, r = 0, s = 32

Number of degrees of freedom : 7
Chi2 statistic for size of jumps : 4.77
p-value of test : 0.69

Normal statistic for number of jumps : -1.58
p-value of test : 0.94

CPU time used : 00:00:00.18

Generator state:
67127040 bits have been read.

HOST = matheus-IdeaPad-3-15ALC6, Linux

ufile_CreateReadBin: /home/matheus/Área de
Trabalho/qanalog_prng-main/qanalog_prng-main/outputs/out_config12_seed3.bin

scomp_LempelZiv test:

N = 1, n = 16777216, r = 0, s = 32, k = 24

Normal statistic : 0.52
p-value of test : 0.30

CPU time used : 00:00:00.41

Generator state:

83904256 bits have been read.

HOST = matheus-IdeaPad-3-15ALC6, Linux

ufile_CreateReadBin: /home/matheus/Área de
Trabalho/qanalog_prng-main/qanalog_prng-main/outputs/out_config12_seed3.bin

sspectral_Fourier1 test:

N = 1, n = 1048576, r = 0, s = 32, k = 20

Normal statistic : 0.47
p-value of test : 0.32

CPU time used : 00:00:00.19

Generator state:

84952832 bits have been read.

HOST = matheus-IdeaPad-3-15ALC6, Linux

ufile_CreateReadBin: /home/matheus/Área de
Trabalho/qanalog_prng-main/qanalog_prng-main/outputs/out_config12_seed3.bin

sspectral_Fourier3 test:

N = 8065, n = 2048, r = 0, s = 32, k = 11

Kolmogorov-Smirnov+ statistic = D+ : 0.048
p-value of test : 0.09

Kolmogorov-Smirnov- statistic = D- : 0.013
p-value of test : 0.83

Anderson-Darling statistic = A2 : 1.31
p-value of test : 0.23

CPU time used : 00:00:00.12

Generator state:

101728032 bits have been read.

HOST = matheus-IdeaPad-3-15ALC6, Linux

ufile_CreateReadBin: /home/matheus/Área de
Trabalho/qanalog_prng-main/qanalog_prng-main/outputs/out_config12_seed3.bin

sstring_LongestHeadRun test:

N = 1, n = 60, r = 0, s = 32, L = 279616

Number of degrees of freedom : 3
Chi-square statistic : 2.59
p-value of test : 0.46

Global longest run of 1 : 23.00
p-value of test : 0.50

CPU time used : 00:00:00.06

Generator state:
118504992 bits have been read.

HOST = matheus-IdeaPad-3-15ALC6, Linux

ufile_CreateReadBin: /home/matheus/Área de
Trabalho/qanalog_prng-main/qanalog_prng-main/outputs/out_config12_seed3.bin

sstring_PeriodsInStrings test:

N = 1, n = 524288, r = 0, s = 31

Number of degrees of freedom : 94
Chi-square statistic : 116.62
p-value of test : 0.06

CPU time used : 00:00:00.02

Generator state:
135282208 bits have been read.

HOST = matheus-IdeaPad-3-15ALC6, Linux

ufile_CreateReadBin: /home/matheus/Área de
Trabalho/qanalog_prng-main/qanalog_prng-main/outputs/out_config12_seed3.bin

sstring_HammingWeight test:

N = 1, n = 524288, r = 0, s = 32, L = 32

Number of degrees of freedom : 22
Chi-square statistic : 17.80
p-value of test : 0.72

CPU time used : 00:00:00.00

Generator state:
152059424 bits have been read.

HOST = matheus-IdeaPad-3-15ALC6, Linux

ufile_CreateReadBin: /home/matheus/Área de
Trabalho/qanalog_prng-main/qanalog_prng-main/outputs/out_config12_seed3.bin

sstring_HammingCorr test:

N = 1, n = 524288, r = 0, s = 32, L = 32

Normal statistic : -1.51
p-value of test : 0.94

CPU time used : 00:00:00.00

Generator state:
168836640 bits have been read.

HOST = matheus-IdeaPad-3-15ALC6, Linux

ufile_CreateReadBin: /home/matheus/Área de
Trabalho/qanalog_prng-main/qanalog_prng-main/outputs/out_config12_seed3.bin

sstring_HammingCorr test:

N = 1, n = 262144, r = 0, s = 32, L = 64

Normal statistic : -0.27
p-value of test : 0.61

CPU time used : 00:00:00.01

Generator state:
185613856 bits have been read.

HOST = matheus-IdeaPad-3-15ALC6, Linux

ufile_CreateReadBin: /home/matheus/Área de
Trabalho/qanalog_prng-main/qanalog_prng-main/outputs/out_config12_seed3.bin

sstring_HammingCorr test:

N = 1, n = 131072, r = 0, s = 32, L = 128

Normal statistic : 1.10
p-value of test : 0.14

CPU time used : 00:00:00.00

Generator state:
202391072 bits have been read.

HOST = matheus-IdeaPad-3-15ALC6, Linux

ufile_CreateReadBin: /home/matheus/Área de
Trabalho/qanalog_prng-main/qanalog_prng-main/outputs/out_config12_seed3.bin

sstring_HammingIndep test:

N = 1, n = 524288, r = 0, s = 32, L = 16, d = 0

Counters with expected numbers >= 10

Number of degrees of freedom : 177
Chi-square statistic : 155.62
p-value of test : 0.87

CPU time used : 00:00:00.00

Generator state:
219168288 bits have been read.

HOST = matheus-IdeaPad-3-15ALC6, Linux

ufile_CreateReadBin: /home/matheus/Área de

Trabalho/qanalog_prng-main/qanalog_prng-main/outputs/out_config12_seed3.bin

sstring_HammingIndep test:

N = 1, n = 262144, r = 0, s = 32, L = 32, d = 0

Counters with expected numbers >= 10

Number of degrees of freedom : 305
Chi-square statistic : 299.04
p-value of test : 0.59

CPU time used : 00:00:00.00

Generator state:

235945504 bits have been read.

HOST = matheus-IdeaPad-3-15ALC6, Linux

ufile_CreateReadBin: /home/matheus/Área de

Trabalho/qanalog_prng-main/qanalog_prng-main/outputs/out_config12_seed3.bin

sstring_HammingIndep test:

N = 1, n = 131072, r = 0, s = 32, L = 64, d = 0

Counters with expected numbers >= 10

Number of degrees of freedom : 489
Chi-square statistic : 541.10
p-value of test : 0.05

CPU time used : 00:00:00.01

Generator state:

252722720 bits have been read.

HOST = matheus-IdeaPad-3-15ALC6, Linux

ufile_CreateReadBin: /home/matheus/Área de

Trabalho/qanalog_prng-main/qanalog_prng-main/outputs/out_config12_seed3.bin

sstring_AutoCor test:

N = 1, n = 16777153, r = 0, s = 32, d = 1

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Normal statistic          :    0.46
p-value of test          :    0.32
-----
```

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CPU time used            : 00:00:00.00
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Generator state:
269499904 bits have been read.
```

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```
HOST = matheus-IdeaPad-3-15ALC6, Linux
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```
ufile_CreateReadBin: /home/matheus/Área de
Trabalho/qanalog_prng-main/qanalog_prng-main/outputs/out_config12_seed3.bin
```

```
sstring_AutoCor test:
```

```
-----
N = 1, n = 16777154, r = 0, s = 32, d = 2
-----
```

```
-----
Normal statistic          :    0.52
p-value of test          :    0.30
-----
```

```
CPU time used            : 00:00:00.01
```

```
Generator state:
286277088 bits have been read.
```

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*****
```

```
HOST = matheus-IdeaPad-3-15ALC6, Linux
```

```
ufile_CreateReadBin: /home/matheus/Área de
Trabalho/qanalog_prng-main/qanalog_prng-main/outputs/out_config12_seed3.bin
```

```
sstring_Run test:
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N = 1, n = 3355443, r = 0, s = 32
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```

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Total number of 1 runs: 3355449
```

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Number of degrees of freedom : 36
Chi2 statistic for number of runs : 21.41
p-value of test : 0.97
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```

```
-----
Total number of bits: 13420928
```

```
Normal statistic for number of bits : -0.16
p-value of test : 0.56
```

CPU time used : 00:00:00.05

Generator state:
299698016 bits have been read.

HOST = matheus-IdeaPad-3-15ALC6, Linux

ufile_CreateReadBin: /home/matheus/Área de
Trabalho/qanalog_prng-main/qanalog_prng-main/outputs/out_config12_seed3.bin

smarsa_MatrixRank test:

N = 1, n = 16384, r = 0, s = 32, L = 32, k = 32

Number of degrees of freedom : 3
Chi-square statistic : 4.31
p-value of test : 0.23

CPU time used : 00:00:00.06

Generator state:
316475232 bits have been read.

HOST = matheus-IdeaPad-3-15ALC6, Linux

ufile_CreateReadBin: /home/matheus/Área de
Trabalho/qanalog_prng-main/qanalog_prng-main/outputs/out_config12_seed3.bin

smarsa_MatrixRank test:

N = 1, n = 163, r = 0, s = 32, L = 320, k = 320

Number of degrees of freedom : 2
Chi-square statistic : 1056.73
p-value of test : eps *****

CPU time used : 00:00:00.07

Generator state:
333166432 bits have been read.

HOST = matheus-IdeaPad-3-15ALC6, Linux

ufile_CreateReadBin: /home/matheus/Área de
Trabalho/qanalog_prng-main/qanalog_prng-main/outputs/out_config12_seed3.bin

swalk_RandomWalk1 test:

N = 1, n = 131072, r = 0, s = 32, L0 = 128, L1 = 128

Test on the values of the Statistic H

Number of degrees of freedom	:	42
ChiSquare statistic	:	55.43
p-value of test	:	0.08

Test on the values of the Statistic M

Number of degrees of freedom	:	42
ChiSquare statistic	:	42.61
p-value of test	:	0.44

Test on the values of the Statistic J

Number of degrees of freedom	:	64
ChiSquare statistic	:	66.33
p-value of test	:	0.40

Test on the values of the Statistic R

Number of degrees of freedom	:	36
ChiSquare statistic	:	26.50
p-value of test	:	0.88

Test on the values of the Statistic C

Number of degrees of freedom	:	22
ChiSquare statistic	:	15.99
p-value of test	:	0.82

CPU time used : 00:00:00.06

Generator state:
349943648 bits have been read.

HOST = matheus-IdeaPad-3-15ALC6, Linux

ufile_CreateReadBin: /home/matheus/Área de
Trabalho/qanalog_prng-main/qanalog_prng-main/outputs/out_config12_seed3.bin

swalk_RandomWalk1 test:

N = 1, n = 16384, r = 0, s = 32, L0 = 1024, L1 = 1024

Test on the values of the Statistic H

Number of degrees of freedom	:	94
ChiSquare statistic	:	136.73
p-value of test	:	2.7e-3

Test on the values of the Statistic M

Number of degrees of freedom	:	94
ChiSquare statistic	:	113.33
p-value of test	:	0.09

Test on the values of the Statistic J

Number of degrees of freedom	:	512
ChiSquare statistic	:	505.54
p-value of test	:	0.57

Test on the values of the Statistic R

Number of degrees of freedom	:	90
ChiSquare statistic	:	72.86
p-value of test	:	0.91

Test on the values of the Statistic C

Number of degrees of freedom	:	50
ChiSquare statistic	:	65.62
p-value of test	:	0.07

CPU time used : 00:00:00.04

Generator state:

366720864 bits have been read.

HOST = matheus-IdeaPad-3-15ALC6, Linux

ufile_CreateReadBin: /home/matheus/Área de
Trabalho/qanalog_prng-main/qanalog_prng-main/outputs/out_config12_seed3.bin

swalk_RandomWalk1 test:

N = 1, n = 1675, r = 0, s = 32, L0 = 10016, L1 = 10016

Test on the values of the Statistic H

Number of degrees of freedom	:	130
ChiSquare statistic	:	143.30
p-value of test	:	0.20

Test on the values of the Statistic M

Number of degrees of freedom	:	131
ChiSquare statistic	:	121.58
p-value of test	:	0.71

Test on the values of the Statistic J

Number of degrees of freedom	:	161
ChiSquare statistic	:	151.65
p-value of test	:	0.69

Test on the values of the Statistic R

Number of degrees of freedom	:	130
ChiSquare statistic	:	159.69
p-value of test	:	0.04

Test on the values of the Statistic C

Number of degrees of freedom	:	89
ChiSquare statistic	:	124.14
p-value of test	:	8.2e-3

CPU time used : 00:00:00.04

Generator state:

383497664 bits have been read.

===== Summary results of Rabbit =====

Version:	TestU01 1.2.3
Generator:	ufile_CreateReadBin
Number of bits:	16777216
Number of statistics:	39

Total CPU time: 00:00:06.10
The following tests gave p-values outside [0.001, 0.9990]:
(eps means a value < 1.0e-300):
(eps1 means a value < 1.0e-15):

Test	p-value
-----	-----
22 MatrixRank, 320 x 320	eps
-----	-----

All other tests were passed